

F-adopt-vs-rewrite

F — Adopt vs Refactor vs Rewrite vs Net-New decision table

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Per-module disposition of the reference BobaLink implementation (44 .ts files under docs/research/browser_extension/Browser Torrent Extension Guide/src/src/) for the Boba browser-extension build. Decisions are evidence-based: every defect is cited to the reference file (and verified by direct read where the extraction’s claim was ambiguous).

Decision vocabulary - **ADOPT** — reuse mostly as-is (rename/port only; no logic change). §11.4.74 reuse. - **REFACTOR** — reuse the structure, fix bounded defects (1-N targeted edits). - **REWRITE** — defects too deep / design wrong; rebuild from the contract, may salvage shape. - **NET-NEW** — does not exist in the reference; must be authored.

Anti-bluff (§11.4): a module is ADOPTed only when reading the source shows it is genuinely sound. Defective code is never ADOPTed just because it exists.

Reference path prefix (note the spaces), abbreviated <REF>/ below:
docs/research/browser_extension/Browser Torrent Extension Guide/src/src/

1. Parser layer

Module/ File	Quality assessment (defect cited or “sound”)	Decision	Required fixes	Risk
		ADOPT		Low

Module/ File	Quality assessment (defect cited or “sound”)	Decision	Required fixes	Risk
parser/ bencode.ts	Sound. Zero-dep Uint8Array bencode; dict keys sorted lexicographically (bencode.ts:139-151) — load-bearing for stable infohash; rejects trailing data, unterminated ints/strings/lists/dicts; bounds-checks byte-strings; binary-vs-utf-8 mode; sha1 via Web Crypto. Strongly unit-tested (bencode.test.ts).		Remove dead-assert in test (bencode.test.ts:156). Add property/fuzz + deep-recursion-limit (chaos) tests per §11.4.85. No source change.	
parser/ magnet.ts	Sound. Repeated-key query parsing (multiple tr/xt preserved), hex+base32 infohash, base32ToHex RFC4648, full param extraction (xt/dn/tr/ws/xl/xs/kt/as/mt), recurse on base32. Fixture-driven tests.	ADOPT	Wire the unused 7-case invalidMagnets negative fixture. Consider BitTorrent v2 (btmh/multihash) — out of scope unless required.	Low
parser/ torrent- file.ts	Sound but DEAD CODE. parseTorrentFile/computeInfohash correct (infohash = sha1(encode(infoDict))), BT v1 canonical) and tested — but never invoked by scanners/adapters/background (sends upload raw File). Re-encode reorders non-sorted info keys (hash-divergence caveat, torrent-file.ts:46-48).	REFACTOR	Source is sound → keep. The fix is wiring : invoke computeInfohash in the scanner/adapters to derive stable ids + local dedup against Boba. §11.4.124 (don’t delete unwired code — wire it).	Med (wiring touches scanner + adapters/contracts)

2. Scanner layer

Module/File	Quality assessment	Decision	Required fixes	Risk
scanner/base.ts	One real defect. Abstract base + shadow-DOM-aware querySelectorAllDeep + hidden-element skip + re-entrancy guard are sound. Defect (verified by read): hashString returns Math.abs(hash).toString(36) + Date.now().toString(36) (base.ts:272) — time-salts the id despite the doc-comment “stable ID”, so the same torrent gets a new id every scan; cross-scan/cross-tab dedup is unreliable.	REFACTOR	Drop the Date.now() salt; derive id from infohash (now available once torrent-file.ts is wired) so ids are stable. One-method change.	Medium correctness dependency
scanner/link-scanner.ts	Sound. Site-specific pass (resolve selectors by domain) + generic a[href] fallback, dedup by normalized href, relative→absolute resolution, sameOrigin flag. Sound.	ADOPT	Reconcile the selector source (see site-db row).	Low
scanner/text-scanner.ts	TreeWalker(SHOW_TEXT) with skip-list + min-length filter, findMagnetUri dedup. Catches forum-pasted magnets.	ADOPT	None.	Low
scanner/orchestrator.ts	Sound. Coordinates scanners + MutationObserver, per-site debounce, cooperative yieldToBrowser() between scanners, relevant-mutation filter, emits scan events, dedups by id into a Map. Initial-state-only tested (orchestrator never actually scans in the reference suite).	ADOPT	Depends on the base.ts stable-id fix for reliable mergeResults dedup. Add real scan + mutation-rescan + debounce tests (the reference does not exercise scanning).	Medium correctness
		REFACTOR		

Module/File	Quality assessment	Decision	Required fixes	Risks
scanner/site-db.ts	Defect: two divergent selector tables. SITES (15 rich entries, used ONLY for per-site debounce) overlaps and disagrees with constants.SITE_SELECTORS (~21 flat entries, used by LinkScanner). The private trackers Boba actually supports (rutracker/iptorrents/kinozal/nnmclub) are split across both.		Merge into ONE authoritative site registry (single source of truth) covering Boba's real trackers (rutracker, kinozal, nnmclub, rutor, IPTorrents). Reconcile against the SQL site_selectors seed (30 rows) too.	Med reco con site SQL

3. API layer

Module/File	Quality assessment	Decision	Required fixes
api/client.ts	Sound HTTP client, WRONG TARGET. TokenBucket rate-limit, AbortController timeout, retryWithBackoff, 429→RateLimitError, content-type branching — all sound. But it speaks ONLY raw qBittorrent WebUI v2 (/api/v2/...) and the default base assumes qBittorrent on :8080. Nothing reaches Boba's real services (7186 proxy / 7187 merge / 7189 jackett). set-cookie SID parse (client.ts:128-166) is browser-fragile.	REFACTOR	Keep the tra (rate-limit/re timeout/erro mapping). Re base URL/ endpoints to real ports an routes; add a endpoint pat distinct from raw qBt path Harden SID handling.
api/auth.ts	Mostly sound, one crypto-coupling defect. Four auth methods + failure backoff + refreshIfNeeded are sound. createCredentialsFromConfig(config, passphrase) (auth.ts:170) decrypts via shared/crypto — its correctness is hostage to the broken passphrase	REFACTOR	Re-wire to th key source (I NEW below) Replace the private-field with a public accessor on t client. Decid

Module/File	Quality assessment	Decision	Required fixes
	provenance (see crypto rows). Reads client["authCookie"] private field (auth.ts:247) — fragile TS escape hatch.		which auth n Boba actually (delegate to jackett vs ray login).
api/ qbittorrent.ts	Sound. Domain wrapper: sendTorrent/sendTorrents (250ms spacing), buildAddOptions maps ServerConfig→add params, never rethrows (normalizes to getUserMessage()).	ADOPT	Inherits the n targeted clie None of its o logic is defeo
api/queue.ts	Defect: never authenticates (verified by read). Enqueue/dequeue/FIFO-eviction/persist sound. But processQueue (queue.ts:170) builds new BobaAPIClient(config.url,...) + adapter and runs the send loop with no login()/authenticate() call — retries against an authed server 403/fail-loop forever.	REFACTOR	Inject auth in processQueue (build+authent the client, or an authentic client/adapte retry/attempt backoff tests (reference on asserts struc
api/health.ts	Sound, wrong port list. Health thresholds, caching, testConnection, autoDiscover. Defect: autoDiscover probes [8080, 7187, 7189] (health.ts:201) — 8080 is wrong for Boba (qBt WebUI is proxied on 7186).	REFACTOR	Change prob set to Boba's map (7186/7187/7

4. Content layer

Module/ File	Quality assessment	Decision	Required fixes	Risk
content/ index.ts	Sound. Wires orchestrator→background (scan-result), message handler (scan-now/get-detected/toggle-selection/get-scan-status). Defect (contract): content get-detected returns {torrents:[...]} while background get-detected returns {data:{result}} — divergent shapes the popup juggles.	REFACTOR	Unify the get-detected response contract across content + background + popup.	Med

Module/ File	Quality assessment	Decision	Required fixes	Risk
content/ scanner.ts	Sound. Thin content-side orchestrator glue (90 lines).	ADOPT	None.	Low
content/ highlight.ts	Sound. Three highlight styles (badge/border/glow), event-bus driven.	ADOPT	None (presentation).	Low
content/ styles.css	Presentation only, .bobalink-* + !important.	ADOPT	Rebrand class prefix if extension is renamed.	Low

5. Entrypoints (background / popup / options)

Module/ File	Quality assessment	Decision	Required fixes	Risk
background/ index.ts	Sound hub, ONE critical defect. Lifecycle, message router, context menus, commands, alarms (keepalive ~20s / health 5min), badge/notifications — all sound. Critical (verified index .ts:409): sendTorrents decrypts creds with empty passphrase "" — can only ever read ""-encrypted bundles, and is mutually incompatible with the options page's "bobalink-	REFACTOR	Replace the "" decrypt call-site with the real key source. Fix :8080 fallback. Add the missing background unit/integration coverage.	HIGH (credential correctness + the unrouted message types decision)

Module/ File	Quality assessment	Decision	Required fixes	Risk
	extension" encryption ⇒ the whole credential path is broken end- to-end. open- dashboard fallback hardcodes :8080. Has zero unit coverage in the reference.			
popup/ popup.ts + popup/ index.html + popup/ styles.css	Sound. Per-tab list, select/ deselect, send, connection- status dot, empty-state scan. Excluded from reference coverage.	ADOPT	Adapt to the unified get- detected contract. Add real UI/UX behavior tests (reference e2e asserts presence only).	Low-Med
options/ options.ts + index.html + styles.css	CRITICAL defect (verified options.ts:327). Form↔config bridge, server modal, test- connection, settings toggles all sound. But every secret is AES- encrypted with the literal fixed passphrase "b obalink- extension" — trivially reversible by anyone with the source	REFACTOR	Remove the fixed passphrase; route encryption through the real key source (NET- NEW). Wire the auto- discover checkboxes to actual config.	HIGH (\$11.4.10 credential leak is a release blocker)

Module/ File	Quality assessment	Decision	Required fixes	Risk
	(\$11.4.10 leak). Auto-discover checkboxes are display-only (real ports hardcoded elsewhere).			

6. Shared libs

Module/ File	Quality assessment	Decision	Required fixes
shared/ constants.ts	Sound values, WRONG PORTS + duplication. Regexes, timeouts, retry/rate config, badge colors, encryption params all sound. Defects: DEFAULT_PORTS.QBITTORRENT=8080 / DEFAULT_URLS.QBITTORRENT wrong (should be 7186); SITE_SELECTORS duplicates site-db.SITES; RATE_LIMIT.MAX_REQUESTS=10 disagrees with DB seed behavior.rate_limit_requests=30.	REFACTOR	Fix qBt port 8080→7186 across host_permissions/CSP/URLs/PORTS/auto-discover. Collapse the duplicate selector table. Pick canonical rate-limit.
shared/ crypto.ts	SOUND PRIMITIVE — extraction's "REWRITE" is too broad (DISAGREE, see §8). Verified by read: PBKDF2 100k iters / AES-256-GCM / fresh 16-byte salt + 12-byte IV per op / 128-bit GCM tag / proper StorageError wrapping / rejects empty plaintext+passphrase. The <i>crypto module itself is correct and reusable</i> . The defect lives entirely in the callers' passphrase (options.ts:327, background:409), not here.	ADOPT (module)	None to the module. The fix is the NET-NEW key source feeding it. Rewrite the test to import this real module (reference test tests a copy).
shared/ errors.ts	Sound. Typed error taxonomy (BobaLinkError + 8 subclasses), getUserMessage, normalizeError, isBobaLinkError.	ADOPT	Add direct error-class unit tests.
		ADOPT	None.

Module/ File	Quality assessment	Decision	Required fixes
shared/ events.ts	Sound. TypedEventEmitter + 13-event EventMap, per-listener try/catch isolation, unsub returns. Thoroughly tested.		
shared/ logger.ts	Sound. Structured leveled logging, timed, createLogger, debug-gating.	ADOPT	None.
shared/ storage.ts	Sound. chrome.storage.local wrapper, NamespacedStorage, change listener, bobalink_-prefixed clear. Note: only the JSON-blob model is implemented here — the parallel SQL-schema (sql.js) model has NO wrapper file.	ADOPT	Decide one persistence source of truth (JSON-blob vs sql.js); the sql.js wrapper is NET-NEW if chosen. Add direct storage unit tests (always mocked away in the reference).
shared/ utils.ts	Sound. debounce/throttle, retryWithBackoff, TokenBucket, processInChunks, formatBytes, url helpers. Note: escapeHtml is DOM-dependent (not service-worker safe); processInChunks defined but unused.	ADOPT	Provide a SW-safe escapeHtml variant for background context.

7. Types

Module/ File	Quality assessment	Decision	Required fixes	Risk
types/ torrent.ts	Sound. MagnetInfo/ TorrentFile/ ParsedTorrent/ DetectedTorrent/ SendResult/ PageScanResult.	ADOPT	None.	Low
types/ config.ts	Sound, with duplication. ServerConfig/ ExtensionConfig/ DEFAULT_CONFIG. Defect: AuthMethod	REFACTOR	De-duplicate AuthMethod (single canonical). Reconcile TS ServerConfig ↔ SQL server_config. Fix default ports	Low

Module/ File	Quality assessment	Decision	Required fixes	Risk
types/ api.ts	duplicated in types/api.ts; ServerConfig (TS) diverges from server_config (SQL) field naming. Sound, declares unused contracts. qBt/ auth/health/ queue/message types. BobaServerInfo/ BobaSearchResponse declared but nothing consumes them (no Boba-API client). MessageType declares 7 values with no handler branch.	REFACTOR	(auto-discovery [7187,7189,8080]→ Boba map). De-dup AuthMethod. Decide which declared-but- unrouted message types to implement vs drop. The Boba- API consumer that uses BobaSearchResponse is NET-NEW.	Low (types) / High (the consumer is net- new)

8. SQL schema

Module/File	Quality assessment	Decision	Required fixes	Risk
sql/schemas.sql (= migrations/ 001_initial.sql)	Sound schema, wrong default + unresolved dual- storage. 9 tables, ~25 indexes, partial indexes, encrypted- credential columns (api_key_encrypted/ password_encrypted), 30 selector seeds, 25 config seeds — well- formed SQLite 3.38+/ sql.js. Defects: seeds qbittorrent_port DEFAULT 8080 + server.base_url=http:// localhost:8080; and the	REFACTOR (if SQL chosen) / drop (if JSON-blob chosen)	First decide the single persistence model. If SQL: fix port seeds 8080→7186, reconcile site-selector seed with the merged site registry, author the sql.js wrapper (NET-NEW).	Med (architect decision gates everything)

Module/File	Quality assessment	Decision	Required fixes	Risk
	runtime has NO sql.js wrapper (only the JSON-blob storage path is implemented) so it is currently unused-by-design.		If JSON-blob: this schema is not adopted.	

9. Build / test configs

Module/File	Quality assessment	Decision	Required fixes
wxt.config.ts	Sound MV3 manifest gen, wrong endpoints. WXT manifest, 6 permissions, CSP, commands, auto-icons. Defects: host_permissions + CSP connect-src include :8080 (should be :7186); icon PNGs referenced but absent.	REFACTOR	Fix 8080→7186 in host_permissions + CSP. Confirm Firefox MV3 module-SW support.
tsconfig.json	Sound. Very strict (all strict flags + noUncheckedIndexedAccess + exactOptionalPropertyTypes).	ADOPT	None.
package.json	Sound, drift. Deps/scripts fine. Defects: HTTPS repo URL (Boba mandates SSH); release.yml references non-existent build:chrome script.	REFACTOR	SSH remote; rebrand name; align scripts; drop CI-only assumptions.
jest.config.ts	Defect: 80% threshold + UI/entrypoints excluded from coverage. collectCoverageFrom excludes popup/options/content/background/**/index.ts — UI/entrypoint logic has ZERO coverage enforcement. Conflicts with §11.4.27.	REWRITE (or replace w/ Vitest)	Decide runner (Vitest recommended). matches Boba frontend/ + the crypto test). Bring UI/entrypoints INTO coverage. raise toward 100%.
playwright.config.ts	Defect: non-functional e2e config. References missing global-setup.ts/global-teardown.ts; dead webServer.url :3000; Firefox project can't load extension via Chromium args;	REWRITE	Author global setup/teardown (build dist, launch persistent Chromium w/ --load-extension --capture-real

Module/File	Quality assessment	Decision	Required fixes
	reporter:github+forbidOnly:CI assume GitHub Actions.		extension id, rewrite chrom extension:// <id>/...). Remove GH- Actions assumptions (Boba manual CI).
.eslintrc.json / .prettierrc	Sound. Type-aware lint (no- explicit-any/no-floating- promises errors), Prettier 100- col.	ADOPT	Align with Bo project lint config.
_locales/en/ messages.json	Sound. 28 i18n keys.	ADOPT	Rebrand string if renamed.
.github/workflows/ ci.yml, release.yml	FORBIDDEN. Boba Hard- Stop: no .github/workflows/ *.yml ever.	drop (do NOT port)	Extract only t <i>commands</i> (li compile/test/ build/zip/ sha256) into manual ci.sh-style script. Never copy the YAM

10. Tests

Module/File	Quality assessment	Decision	Required fixes	Risk
bencode.test.ts	Sound; dead-assert line 156.	ADOPT	Remove dead assert; add fuzz/chaos.	Low
magnet.test.ts	Sound, fixture-driven; invalidMagnets (7) unused.	ADOPT	Wire negative fixtures.	Low
torrent-file.test.ts	Sound (infohash determinism, single/multi parse).	ADOPT	None.	Low
api-client.test.ts	Partial BLUFF. Auth block ends expect(true).toBe(true) (no observable check); 429 test asserts ServerError despite name saying RateLimitError.	REFACTOR	Replace the bluff Auth assertions with real cookie-header propagation checks; fix	Med

Module/File	Quality assessment	Decision	Required fixes	Risk
queue.test.ts	Partial BLUFF. Auto-processing block ends <code>expect(true).toBe(true)</code> ; <code>processQueue</code> only asserts <i>structure</i> , not success/failure.	REFACTOR	the 429 assertion. Assert real retry/attempts/outcomes; remove no-ops.	Med
scanner.test.ts	Weak. Event-emitter block sound; but orchestrator block asserts initial-state only (never scans); DOM block tests <code>querySelectorAll</code> , NOT the production scanner.	REFACTOR	Drive the real orchestrator over real DOM; assert detected torrents + mutation rescans.	Med
crypto.test.ts	BLUFF (§11.4.1). 44 thorough tests — but against an inline re-implemented copy , never <code>src/shared/crypto.ts</code> . Cannot fail against a no-op stub of the real module. Also imports <code>vitest</code> while the other 6 files use Jest (mixed runners).	REWRITE	Re-point every test at the production <code>shared/crypto</code> module; standardize runner.	Med (i.e. the most-thorough file but proves nothing as-is)
chrome-mock.ts / setup.ts	Mock <code>infra</code> ; happy-path only (no error injection), no cookies/scripting/sync; <code>setup.ts:14</code> wrong path (type-only).	REFACTOR	Add error-injecting variant for chaos/stress; fix the stale path.	Low
content.spec.ts / options.spec.ts / popup.spec.ts (e2e)	NON-FUNCTIONAL. Hardcoded <code>chrome-extension://test-id/...</code> ; no fixture loads the extension or resolves the real id; content spec asserts raw DOM, not the injected content script. Asserts presence/visibility, never user outcomes.	REWRITE	Real extension-loading fixture + user-observable assertions (torrent rendered, server persisted, send→appears in qBittorrent).	High (load-bearing for §11 anti-bl e2e)

11. NET-NEW (does not exist in the reference — must be authored)

Item	Why net-new	Risk
Real key source for credential encryption	crypto.ts is a sound primitive but has NO real passphrase provenance — callers use a fixed string + an empty string. Need a master-passphrase prompt, OS-keychain, OR delegation to boba-jackett/proxy (which already own encrypted creds at /config/boba.db). §11.4.10 blocker.	HIGH
Real Boba service integration (7186/7187/7189)	Reference speaks ONLY raw qBittorrent v2; BobaServerInfo/ BobaSearchResponse types are declared but nothing consumes them. The merge-search API (7187), tracker-cookie download path (7186 proxy), and boba-jackett (7189) clients do not exist.	HIGH
chrome.tabGroups tab-group batching	Verified: the reference has NO chrome.tabGroups usage. The only “batching” is intra-tab yieldToBrowser() cooperative yielding + per-tab tabTorrents map + an unused processInChunks. Tab-group batch behavior is fresh design.	Med

Item	Why net-new	Risk
Icon PNG rasterization (16/32/48/128)	Verified: src/assets/ contains icon.svg only — zero .png files. Code/manifest reference rasterized PNGs that must be produced at build (WXT auto-icons from the SVG).	Low
sql.js storage wrapper (only if SQL persistence chosen)	The 9-table schema exists but NO runtime wrapper implements it; only the JSON-blob storage.ts path is live.	Med
Playwright global-setup.ts / global-teardown.ts	Referenced by config, absent from tree. Must build dist, launch persistent Chromium --load-extension, capture real extension id.	Med
Missing test TYPES (§11.4.27/ §11.4.85): integration (real cross-module → live 7186/7187), functional e2e, automation/ HelixQA, security/pen (credential-leak, XSS via magnet dn, CSP, permission-scope), DDoS/load, scaling, chaos (storage corruption / network fault / SW termination), stress, performance, benchmark, UI/UX, Challenges	Reference covers only unit + e2e (and those have bluffs). 13 of 15 mandated types absent.	HIGH (volume)

Reuse estimate

Counting the **44 .ts source files** plus the SQL schema and the build/test configs:

Disposition	Approx. count	Examples
ADOPT (reuse ~as-is)	~17	bencode, magnet, link/text-scanner, content/scanner+highlight, qbittorrent adapter, crypto module, errors, events, logger, storage, utils, types/torrent, eslint/prettier/tsconfig, i18n
REFACTOR (bounded fixes)	~18	torrent-file (wire), base (un-salt id), site-db (merge tables), client (re-target), auth, queue (auth), health (ports), content/index (contract), background, options, popup, constants (ports), types/config, types/api, wxt.config, package.json, chrome-mock, sql schema
REWRITE	~7	jest.config (runner + coverage), playwright.config, crypto.test, api-client.test (bluff), queue.test (bluff), scanner.test, all 3 e2e specs
NET-NEW	7+ areas	key source, Boba-service clients, tab-groups, icon PNGs, sql.js wrapper, e2e global setup, 13 missing test types
DROP	2	both .github/workflows/*.yaml

Disposition	Approx. count	Examples
		(constitution hard-stop)

Headline: roughly 80% of the reference's logic/structure is reusable (ADOPT + REFACTOR $\approx$ 35 of $\sim$ 44 source-bearing files — the parsers, scanners, shared libs, types, and the API transport are sound), **$\sim$ 20% must be rewritten** (the test suite's bluffs/non-functional e2e + the runner/coverage configs), **and the highest-value work is NET-NEW integration** (real Boba endpoints + a real key source) that the reference never attempted. The reference is a strong *skeleton with sound primitives* whose two structural gaps — credential key-management and Boba-service integration — are exactly the parts it left as placeholders.

Top 5 highest-risk rewrites

- 1. Real key source for credential encryption (NET-NEW).**
The sound crypto primitive is currently fed a fixed ("bobalink-extension", options.ts:327) and an empty ("", background:409) passphrase that are even mutually incompatible — the credential path is broken end-to-end and is a §11.4.10 leak (release blocker). Touches options + background + auth.
- 2. Boba-service integration / api/client.ts re-target (REFACTOR→effectively rewrite of the endpoint layer).**
The client speaks only raw qBittorrent v2 on :8080; nothing reaches 7186/7187/7189. Re-targeting + adding the Boba-API consumer (BobaSearchResponse) is the load-bearing integration, and the SID/cookie + CORS/host-permission story is fragile.
- 3. Functional e2e suite (REWRITE all 3 specs + author global-setup/teardown).** Every reference e2e is non-functional (hardcoded chrome-extension://test-id, no extension load, presence-only asserts). §11.4 anti-bluff demands real extension load + user-observable outcomes — this is from-scratch.
- 4. crypto.test.ts (REWRITE).** The single most thorough test file (44 cases) is a §11.4.1 bluff — it tests an inline copy, never the production module, so it passes against a no-op stub. Must be re-pointed at shared/crypto and the runner standardized (mixed Jest/Vitest).
- 5. Test runner + coverage config (jest.config REWRITE).**
80% threshold with UI/entrypoints excluded conflicts with §11.4.27; the runner decision (Vitest vs Jest) is foundational and gates every other test file's port. (Close behind: queue.ts auth-gap fix + the base.ts time-salted-id fix, both correctness-critical for offline retries and dedup.)

Disagreements with the extraction's defect claims (re-verified by reading source)

1. `crypto.ts` → the extraction tags it “ **REWRITE**”; I **DISAGREE** — the **MODULE** is **ADOPT**, the **KEY SOURCE** is **NET-NEW**. Direct read of `shared/crypto.ts` (lines 42-228) shows the encrypt/decrypt machinery is genuinely sound: PBKDF2 100k iterations, AES-256-GCM, a fresh 16-byte random salt + 12-byte IV per operation, a 128-bit GCM auth tag, and correct `StorageError` wrapping that rejects empty plaintext/passphrase. **No defect exists inside this file.** The defects the extraction cites — fixed passphrase “bobalink-extension” (verified `options.ts:327`) and empty passphrase “” (verified `background/index.ts:409`) — live entirely in the *callers*, not in the `crypto` module. So the correct disposition is: **ADOPT the crypto primitive unchanged; author the real key source as NET-NEW; REFACTOR the two call-sites.** Rewriting the `crypto` module itself would be discarding sound, reusable work (§11.4.74). The extraction’s “ `crypto` ... → **REWRITE**” headline conflates the module with its misuse.
2. **Minor: the extraction calls `base.ts` id generation a generic “unstable / time-salted” concern; verified it is a one-line REFACTOR, not a rewrite.** `base.ts:272` literally appends `Date.now().toString(36)` to the hash; removing that suffix and sourcing the id from the (now-wireable) `infohash` is a bounded fix to a single private method — the rest of `base.ts` (shadow-DOM traversal, hidden-element filtering, re-entrancy guard) is sound and ADOPT-grade.

All other extraction defect claims I sampled were verified accurate: the `queue` never authenticates (`queue.ts:170` builds+runs with no login), the `.torrent` parser is genuinely never invoked, the two selector tables genuinely diverge, the icon PNGs are genuinely absent (only `icon.svg`), and the two passphrase call-sites are genuinely incompatible.